

Parallelized MC model for Langmuir-Hinshelwood mechanisms of oxygen on carbon surfaces

Atomistic Scale Simulations

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Project Aim

- Write a Monte Carlo code that simulates a given number of competing Langmuir-Hinshelwood mechanisms for carbon oxidation by atomic oxygen.
- Predict equilibrium surface coverage of mechanisms with temperature and particle flux.
- Predict the oxidation probability of carbon. Which is the carbon product flux out of the surface and oxygen flux to the surface.
- Implement FPGA parallelism to minimize computer run times.

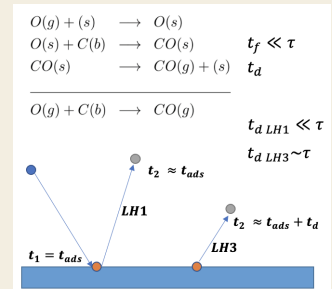
Carbon Oxidation

Type	Mechanisms	Reaction	Rate constant (k)
Adsorption	Adsorption	$O(g) + (s) \rightarrow O(ads)$	$\frac{1}{\phi} + \frac{1}{2} \sqrt{\frac{8k_B T}{\pi m}} \approx 0.85$
Adsorption-mediated GS reactions	LH3 O[a] formation	$O(ads) \rightarrow O[a](s)$	1
	LH3 CO[s] formation	$O(ads) + C(b) + O'(ads) \rightarrow CO[a](s) + O'(ads)$	$\frac{1}{\phi} + 153.03 \exp(-\frac{4472.8}{T_s})$
	LH3 CO[b] formation	$O(ads) + C(b) + O'(ads) \rightarrow CO[b](s) + O'(ads)$	$\frac{1}{\phi} + 71.201 \exp(-\frac{1101.2}{T_s})$
	LH1 O formation	$O(ads) \rightarrow O(TD)(g) + (s)$	$20.879 \exp(-\frac{2489.3}{T_s})$
	LH1 CO formation	$O(ads) + C(b) + O'(ads) \rightarrow CO(g) + (s) + O'(ads)$	$\frac{1}{\phi} + 1574.9 \exp(-\frac{6280.8}{T_s})$
PS reactions	LH1 CO ₂ formation	$O(ads) + O(s) + C(b) + 4O'(ads) \rightarrow CO_2(g) + 2(s) + 4O'(ads)$	$\frac{1}{2\phi} + 536.30 \exp(-\frac{650.63}{T_s})$
	LH3 O[a] desorption	$O[a](s) \rightarrow O(g) + (s)$	$0.050457 T^2 \exp(-\frac{3177.3}{T_s})$
	LH3 CO[s] desorption	$CO[a](s) \rightarrow CO(g) + (s)$	$4485.5 \exp(-\frac{1261.4}{T_s})$
	LH3 CO[b] desorption	$CO[b](s) \rightarrow CO(g) + (s)$	$1.2194 \exp(-\frac{2261.6}{T_s})$

Experiments show carbon oxidation occur with a competing LH 1 and LH 3 mechanisms [1].

LH mechanisms have three major steps [2]:

1. Adsorption
2. Formation
3. Desorption



Program overview: Inputs

- Lattice of given size in two dimensions (X,Y) are created to represent the surface and available adsorption sites.
- Inputs by the user:
 1. X and Y
 2. Temperature (T)
 3. Timestep size (s)
 4. Number of timesteps executed
 5. Number of atoms hitting the surface per timestep
- Program parameters (surface sites, reaction rates etc.) are initialized in a constructor when solver created

```
solver_inputs inputs;  
inputs.temperature = 1000; //K  
inputs.surface_size_X = 100;  
inputs.surface_size_Y = 100;  
inputs.particle_flux = 40;  
inputs.time_step_size = 1e-6;  
inputs.time_step_no = 2000;  
  
Solver solver(inputs);  
solver.execute();
```

Figure: Inputs are given to solver as a struct

Program overview: General Steps

Solver Executor has 4 main functions:

1. `applyGasCollisions()`: Represent flux and perform adsorption/formation reactions
2. `surfaceStep()`: Perform Desorption Reactions
3. `deleteParticles()`: Delete desorbed particles
4. `recordData()`: Write surface coverage and carbon product flux

```
bool Solver::execute() {  
    for (int ti =0; ti<total_time; ti++){  
        surfaceStep();  
        applyGasCollisions();  
        deleteParticles();  
        recordData();  
        std::cout<<"Timestep: "<<ti<<std::endl;  
    }  
    return true;  
}
```

Figure: execute function

Adsorption Phase Algorithm

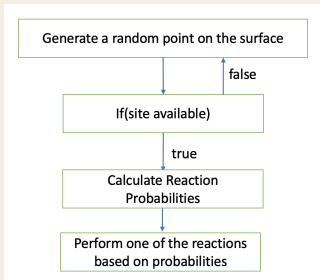


Figure: Adsorption Algorithm

- For reactions that include slow desorption:
- Particle of that adsorbate type created on that location and surface site marked as occupied with selected type.
- Adsorbates desorption time is estimated and time counters (τ) set to zero [4]

$$t_{des} = -\ln(U)/k_{des} \quad (1)$$

Desorption Algorithm

For every adsorbed particle

1. Time counters updated,
 $\tau + = dt$
2. If($\tau > t_{des}$): Particle desorbed
3. Surface is emptied
4. Particle marked for deletion
to free the memory

- Number of adsorbed particles and carbon removal counts are updated with every reaction
- Surface Coverage and carbon removal rates per timestep pushed back to a vector at the end of every time step for post processing.

Model Check: Langmurian Adsorption-Desorption

ODE for Langmurian Adsorption [5]:

$$\frac{d\theta}{dt} = r_A(1 - \theta) - r_D\theta \quad (2)$$

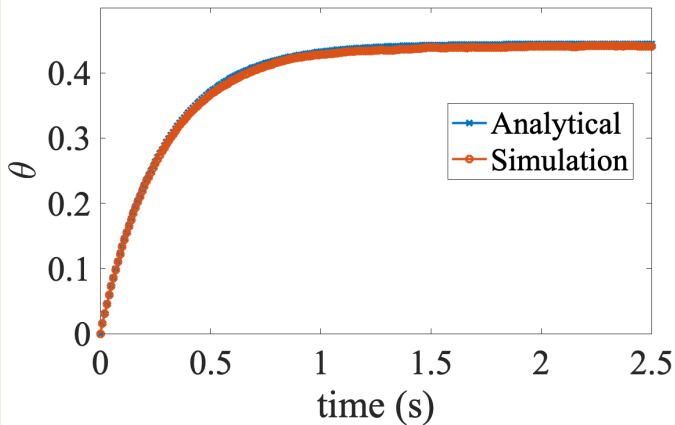
Solution is:

$$\theta(t) = \frac{r_A}{r_A + r_D} (1 - \exp(-(r_A + r_D)t)) \quad (3)$$

$$r_A = S_0(F/B) \quad (4)$$

F = flux, B = Site Density

$$r_A = 1.6, r_D = 2$$



Carbon Oxidation Cases

6 Reactions are taken into account

4 Gas-Surface Reactions:

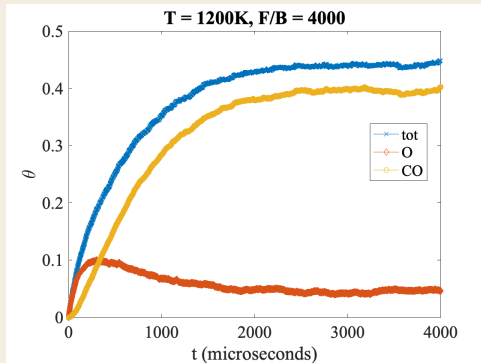
1. LH3 O formation, $k = 1$
2. LH3 CO formation, $k = \theta 153.0 \exp(-4172.8/T)$
3. LH1 O formation/desorption, $k = 20.9 \exp(-2449.3/T)$
4. LH1 CO formation/desorption, $k = \theta 1574.9 \exp(-6240.0/T)$

2 Desorption Reactions:

1. LH3 O desorption, $k = 0.05T^2 \exp(-3177.2/T)$
2. LH3 CO desorption, $k = 4485.5 \exp(-1581.4/T)$

Results

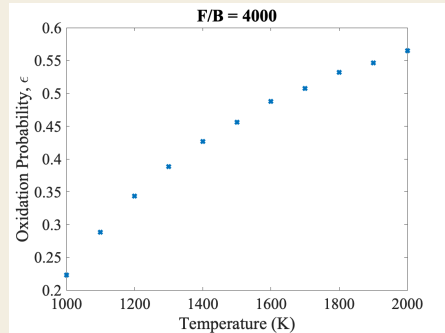
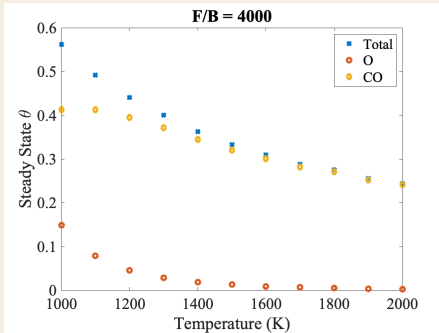
Steady State surface coverage and oxidation probabilities are computed with ranging temperature and flux conditions



Results show non-Arrhenius behavior of oxidation that depends on flux and temperature

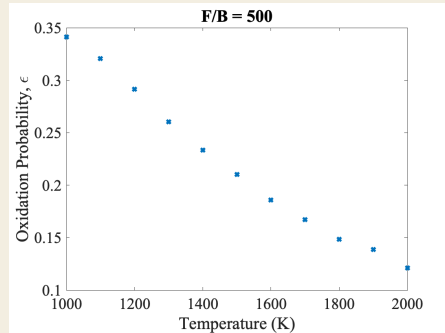
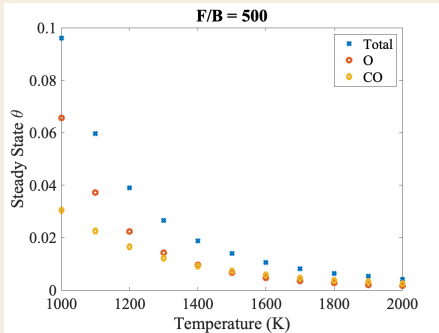
High Flux

Rate Limited, oxidation probability increases with temperature



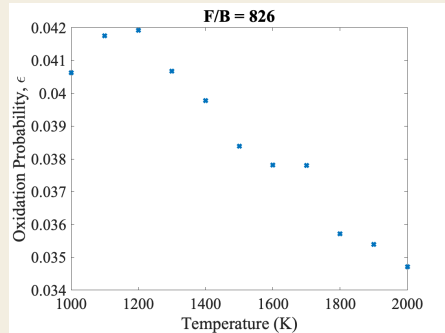
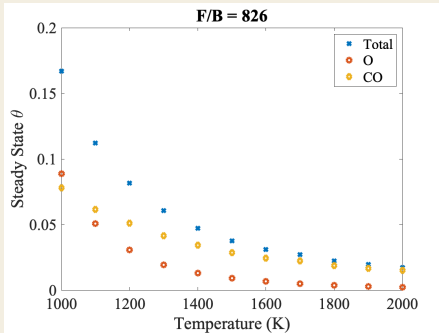
Low Flux

Surface Coverage Limited, oxidation probability decreases with temperature



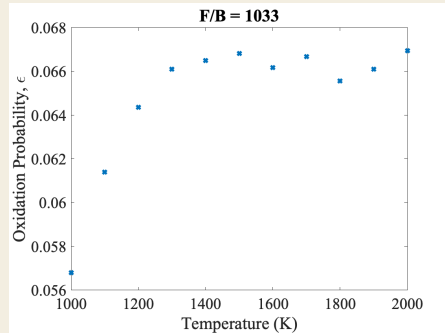
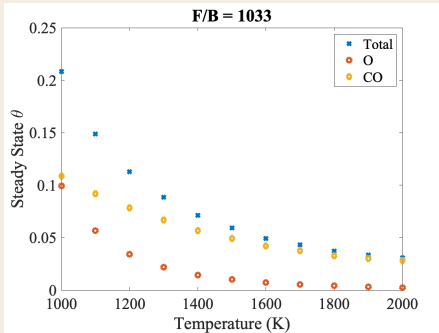
Mid Flux

Competition between surface coverage and rates, maximums can occur with temperature



Mid Flux

Competition between surface coverage and rates, maximums can occur with temperature

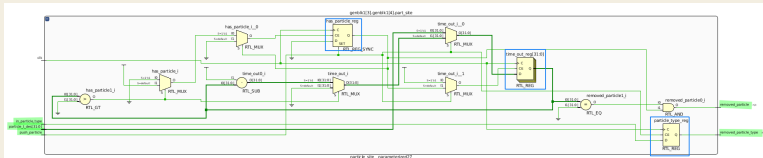


Acceleration on an FPGA

- FPGAs (Field Programmable Gate Arrays) are programmable circuit boards
- Allows for extremely parallel processing
- Running on "bare metal" removes a lot of overhead such as memory speed, operating system overhead, etc.
- Limited in scale and resources
- Previous works have seen upwards of 10^4 times performance gains[\[3\]](#)
- Will attempt to parallelize the desorption, deletion, and parts of data collection

Grid sites

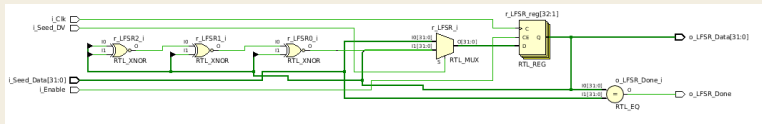
- Create grid of 'sites'
- Each site keeps track of its own data
- Each site can independently update its self in parallel



- Contains data storage for particle type, if there is a particle, and diffusion timer (Blue)
- Inputs if a new particle should be created, outputs if the particle in this site was removed

Random Numbers

- Use Linear Feedback Shift Registers (LFSR)
- Move all bits in binary number to the right by 1
- Replace upper most bit with value calculated based on current value



- Generates uniform random numbers $\in [0, 2^{32} - 1]$
- Use lookup tables to find log of random numbers

Performance and Resource Usage

Resource	Utilization	Available	Utilization %
LUT	76	53200	0.14
FF	34	106400	0.03

Figure: Resource usage for single particle

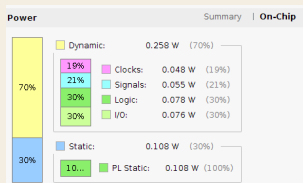


Figure: Power Usage for 16x16 grid

Resource	Utilization	Available	Utilization %
LUT	12758	53200	23.98
FF	9122	106400	8.57

Figure: Resource for 16x16 grid

Total Delay	Logic Delay	Net Delay
14.256	3.752	10.504

Figure: Critical Path for 16x16 grid

At an 15ns delay, this would run at about 66MHz, and would perform one time step in about 150ns. The software implementation took about 30 μ s per step. This gives a speedup of about 200.

Bibliography

- [1] Swaminathan-Gopalan, et al. (2018). Development and validation of a finite-rate model for carbon oxidation by atomic oxygen. Carbon, 137, 313–332. <https://doi.org/10.1016/j.carbon.2018.04.088>
- [2] Swaminathan-Gopalan et al. (2018). Development of a detailed surface chemistry framework in DSMC. AIAA Aerospace Sciences Meeting, 2018, 210059. <https://doi.org/10.2514/6.2018-0494>
- [3] F. Ortega-Zamorano et al. (2016). "FPGA Hardware Acceleration of Monte Carlo Simulations for the Ising Model," in IEEE Transactions on Parallel and Distributed Systems, vol. 27, no. 9, pp. 2618-2627, 1 Sept. 2016, doi: 10.1109/TPDS.2015.2505725.
- [4] Molchanova et al.(2014). "A detailed DSMC surface chemistry model," in AIP Conference Proceedings, vol. 1628, Feb 2015, pp. 131-138, doi:10.1063/1.4902584.
- [5] Masel, Richard (1996). Principles of Adsorption and Reaction on Solid Surfaces. Wiley Interscience. p. 240. ISBN 978-0-471-30392-3.